

Anay Apte

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EDUCATION

University of Pennsylvania

B.S.E. in Computer Science

Philadelphia, PA

Aug. 2025 – May 2028

- **GPA:** 3.6/4.0
- **Relevant Coursework:** Programming Languages & Techniques, Data Structures & Algorithms, Designing LLM Agents, Language Modeling From Scratch (Stanford University), Discrete Mathematics, Probability, Multivariate Calculus, Linear Algebra

EXPERIENCE

Software Engineering Intern

Eigenvector LLC.

May 2024 – June 2025

Naperville, IL

- Built full-stack authentication system using SvelteKit, Drizzle ORM, and PostgreSQL supporting dual account roles (teacher/student), secured with Cloudflare firewall on sign-in flows
- Implemented VM deployment interface enabling real-time provisioning of virtual machines, demoed at **SkillsUSA 2024 Nationals**
- Integrated Stripe payment system into the platform to handle subscription and billing for classroom VM access

Software Engineering Intern

Up Cancer

June 2026 — Present

Atlanta, GA

- Architected a single-table AWS DynamoDB schema for Hatching Sparrow, a full-stack CRM built with Next.js and Express.js, deriving access patterns from an existing frontend codebase without backend documentation.
- Designed PK/SK structure, access patterns, and GSI strategy across Users and Contacts entities, creating a scalable foundation extensible to **20+ planned features** without re-architecting.

Project Lead

Penn Spark

Jan. 2026 – Present

Philadelphia, PA

- Led a cross-functional team of **4 developers** and **3 designers** building SmartSketch, presenting the final product at Penn Spark Demo Day to an audience of **100+ attendees**
- Coordinated weekly sprints and team meetings, maintaining alignment across engineering and design while reporting progress in weekly cross-team PL syncs

Director of Marketing & Singer

Penn Masala (World's First South Asian A Capella Group)

Sep. 2025 – Present

Philadelphia, PA

- Member of the world's first South Asian a cappella group with **400K+ followers** and content exceeding **10M+ views**, with group credits including performances at the **White House** and **2024 Paris Olympics**

PROJECTS

Penn Academic Co-Pilot | *Python, FastAPI, PDFPlumber, Pytest, Git, OpenAI API (GPT-4)*

- Engineered an OpenAI GPT-4 agent with **6 custom tools**, routing live student queries against real Penn course data, correctly blocking ineligible enrollments and resolving multi-constraint scheduling
- Designed a **7-stage guardrail system** mitigating **10+ threat vectors** including prompt injection, hallucinated course recommendations, and state corruption, validated across **3 end-to-end test cases** with unit-tested components

SmartSketch | *TypeScript, Next.js, React, Tailwind CSS, Supabase, Livekit, Gemini API, Deepgram*

- Built a real-time audio pipeline integrating LiveKit, Deepgram, and Gemini processing live speech into structured mindmaps with incremental React Flow updates, validated across **3 live sessions**
- Developed a full-stack Next.js/TypeScript application with live session recording, transcript-grounded AI chat, and an interactive mindmap, with Supabase persisting transcripts, serialized graph data, and audio recordings

TECHNICAL SKILLS

Languages: Java, Python, C++, OCaml, SQL (Postgres), JavaScript, HTML/CSS, LaTeX

Frameworks/Tools: AWS, Git, Flask, React / Native, Node.js, JUnit, FastAPI, Matplotlib, Claude, Gemini, Supabase,

Libraries & APIs: Git, Docker, TravisCI, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse

Activities: Penn Spark, Penn Masala

Hobbies: Weightlifting, Basketball, Tennis, Content Creation, Music Production, Cooking, Fashion